

P-102

CENTRIFUGAL PROCESS PUMP

Operation, Maintenance & Troubleshooting Manual

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IMPORTANT

This is a fictional, synthetic manual created for an SIH prototype/demo. It does not represent an actual MRPL/Oil & Gas facility manual and must not be used for real plant operation.

Prepared for: Sovereign On-Premise Agentic AI Workbench — Demonstration
Knowledge Base

1. Document Control

This document defines fictional operating, inspection, maintenance, and troubleshooting information for a simulated centrifugal process pump identified as P-102.

Field	Value
Equipment Tag	P-102
Equipment Type	Horizontal centrifugal process pump
Service	Simulated process-liquid transfer
Manual ID	P102-O&M-001
Revision	1.0
Status	Synthetic / Demonstration Only
Intended Use	RAG, agent, search and citation testing

Safety classification: **Training / prototype data only.** Real-world plant limits must always come from approved site documentation and qualified personnel.

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3. Equipment Overview

P-102 is a fictional horizontal centrifugal pump used in this demonstration to transfer process liquid from an upstream vessel to a downstream header. The pump is represented as a motor-driven unit with suction and discharge piping, isolation valves, a mechanical seal, bearings, and a local monitoring point.

The simulated pump is intentionally described with enough structured information to allow an AI system to answer questions from the manual and cite the relevant section.

3.1 Intended Function

Provide continuous liquid transfer while maintaining the required discharge pressure and flow within the defined simulated operating envelope.

3.2 Identification

Equipment tag: **P-102**. Driver: **M-102**. Suction line: **S-102**. Discharge line: **D-102**. The tag names are fictional and are used only for prototype testing.

4. Technical Specifications

Parameter	Nominal / Limit
Pump type	Horizontal end-suction centrifugal
Rated flow	120 m³/h
Normal flow range	90–120 m³/h
Rated differential pressure	4.0 bar
Normal discharge pressure	3.5–4.2 bar
Simulated high discharge alarm	4.8 bar
Simulated low discharge alarm	2.5 bar
Normal bearing temperature	45–75 °C
High bearing temperature alarm	85 °C
Maximum simulated bearing temperature	95 °C
Motor rating	30 kW
Nominal motor speed	2,950 rpm
Suction connection	DN100 (simulated)
Discharge connection	DN80 (simulated)

Note: All values in this section are fictional demonstration values. They are not suitable for real equipment.

5. Main Components

5.1 Pump Casing — Contains the process fluid and guides flow through the pump.

5.2 Impeller — Rotating hydraulic component that adds energy to the fluid.

5.3 Mechanical Seal — Limits process-liquid leakage along the rotating shaft.

5.4 Bearings — Support the rotating shaft and require appropriate lubrication/condition monitoring.

5.5 Motor M-102 — Simulated electric driver connected to the pump through a coupling.

5.6 Suction and Discharge Valves — Used to control and isolate the pump in accordance with the approved operating procedure.

5.7 Local Instrumentation — Simulated pressure, temperature, vibration, and motor-current readings.

6. Operating Conditions

The following conditions are the normal simulated envelope used by the SIH demonstration.

Measurement	Normal	Attention / Alarm
Flow	90–120 m³/h	<80 or >130 m³/h
Discharge pressure	3.5–4.2 bar	≥4.8 bar high alarm; ≤2.5 bar low alarm
Bearing temperature	45–75 °C	≥85 °C alarm
Vibration	≤4.5 mm/s RMS	≥7.1 mm/s RMS
Motor current	≤52 A	≥58 A

If any safety-critical or abnormal condition occurs, stop relying on automated recommendations and follow the approved site procedure and qualified human instructions.

7. Startup Procedure

For this fictional training scenario, the startup sequence is:

- 1. Confirm that the pump, motor, coupling guard, and surrounding area are visually acceptable.
- 2. Confirm adequate suction-side liquid availability.
- 3. Confirm the suction path is available according to the approved procedure.
- 4. Verify the discharge path is configured for startup.
- 5. Start motor M-102 using the simulated control interface.
- 6. Observe discharge pressure, flow, bearing temperature, vibration, and motor current.
- 7. Record initial readings in the operating log.
- 8. If abnormal readings persist, stop the simulated operation and investigate.

8. Normal Operation

During operation, the operator should monitor flow, discharge pressure, bearing temperature, vibration, and motor current. A stable pump should remain inside the normal operating envelope listed in Section 6.

Sudden changes in pressure, flow, vibration, temperature, or current should be treated as an abnormal condition requiring investigation rather than ignored.

9. Shutdown Procedure

For the demonstration scenario: reduce the simulated operating load as required, stop motor M-102 through the normal control sequence, verify that the pump has stopped, and record final readings. Isolation or maintenance activities require an approved site procedure and qualified personnel.

10. Troubleshooting

Observed condition	Possible cause	Recommended investigation
High discharge pressure (≥4.8 bar)	Restricted/blocked discharge path	Check discharge line and valve configuration.
Low discharge pressure (≤2.5 bar)	Low suction availability or insufficient pump performance	Check suction condition, flow, and pump status.
Low flow (<80 m³/h)	Suction restriction, discharge restriction, or pump speed reduction	Compare suction and discharge readings and inspect relevant piping.
High bearing temperature (≥85 °C)	Lubrication issue, excessive load, or bearing malfunction	Check lubrication status and inspect bearing condition.
High vibration (≥7.1 mm/s RMS)	Misalignment, imbalance, looseness, or bearing issue	Check coupling/alignment and inspect rotating equipment.
High motor current (≥58 A)	Excessive hydraulic load or mechanical resistance	Compare flow/pressure and inspect for abnormal mechanical load.
Seal leakage	Seal wear or abnormal operating condition	Stop and escalate according to approved maintenance procedure.
Pump does not start	Control/interlock condition or motor fault	Check simulated alarm/interlock status; qualified personnel investigate electrical system.

Troubleshooting principle: first identify the abnormal measurement, then cross-check related measurements and documented procedures. Do not infer a cause when the available evidence is insufficient.

11. Preventive Maintenance

Frequency	Inspection / Task	Record
Each shift	Visual condition; leaks; unusual noise; pressure/flow readings	Operating log
Weekly	Review vibration and bearing-temperature trends	Condition-monitoring log
Monthly	Inspect coupling/guard condition and check for abnormal trends	Maintenance checklist
Quarterly	Review lubrication condition and mechanical condition	Maintenance record
As required	Investigate alarms, abnormal vibration, seal leakage, or repeated trips	Work order / report

12. Safety & Emergency Guidance

This section is intentionally high-level because this PDF is a software-demo knowledge source, not a real maintenance instruction.

- Never bypass a safety interlock for troubleshooting.
- Safety-critical actions require qualified personnel and approved procedures.
- Treat abnormal pressure, temperature, vibration, leakage, or electrical indications as reasons for controlled investigation.
- Do not open, isolate, dismantle, or repair process equipment based solely on an AI response.
- For an emergency, follow the facility emergency plan and contact the responsible control-room/maintenance personnel.

Human-in-the-loop rule: the AI may summarize evidence and suggest documented checks, but it must not independently authorize a safety-critical physical action.

13. Alarm Reference

Alarm ID	Condition	Priority for demo	Suggested AI response
P102-P-HIGH	Discharge pressure ≥ 4.8 bar	High	Check documented high-pressure troubleshooting; inspect dis
P102-P-LOW	Discharge pressure ≤ 2.5 bar	High	Check suction condition and pump/process status.
P102-T-HIGH	Bearing temperature ≥ 85 °C	High	Check temperature trend and lubrication/mechanical condition
P102-V-HIGH	Vibration ≥ 7.1 mm/s RMS	High	Investigate alignment, imbalance, looseness, or bearing cond
P102-I-HIGH	Motor current ≥ 58 A	Medium/High	Cross-check pressure, flow, and mechanical load.

14. Inspection Checklist

A prototype agent can convert this checklist into a structured inspection task:

Item	Expected state
Pump identification	P-102 confirmed
Motor	M-102 available / no active simulated fault
Suction condition	Adequate process availability
Discharge path	Configured according to approved procedure
Discharge pressure	3.5–4.2 bar normal
Flow	90–120 m³/h normal
Bearing temperature	45–75 °C normal
Vibration	≤ 4.5 mm/s RMS normal
Motor current	≤ 52 A normal
Leakage	No abnormal seal leakage

15. Agent / RAG Test Questions

These questions are designed to test whether your SIH system can retrieve the correct information from the manual and provide a source citation.

1. What is the normal discharge pressure of P-102?
2. At what discharge pressure does the high-pressure alarm activate?
3. What should be investigated if P-102 has high discharge pressure?
4. What is the normal flow range?

5. What is the high bearing-temperature alarm limit?
6. What are possible causes of high vibration?
7. What should be checked when motor current is high?
8. What measurements should the operator monitor during normal operation?
9. What does alarm P102-V-HIGH indicate?
10. If the manual does not contain an answer, what should the AI say?

Expected RAG behavior: The agent should answer from retrieved evidence, identify the source document/section, and say “insufficient evidence” when the manual does not support a conclusion.

END OF SYNTHETIC MANUAL

For SIH prototype demonstration only. Replace synthetic values with authorized data only if the problem owner provides it and its use is permitted.